



B(4th Sm.)-Computer Sc.-G/(GE/CC-4)/CBCS

2024

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-4

(Operating Systems)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer *any five* questions :

2×5

- (a) Define system software.
- (b) What do you understand by kernel mode of operating system?
- (c) Differentiate between preemptive and non-preemptive scheduling algorithms.
- (d) What is starvation?
- (e) What is race condition in concurrent processes?
- (f) State the advantages of spooling.
- (g) State difference between physical and logical address space.
- (h) Name any two disk scheduling strategies.

2. (a) Explain real time operating systems and its applications.

(b) State differences between multiprogramming and multiprocessing operating systems.

(c) Write a note on Process Control Block.

3+3+4

3. (a) Explain the four necessary conditions for deadlock.

(b) Explain the critical section problem.

(c) Write the Banker's safety algorithm.

2+2+6

Please Turn Over

4. (a) Consider the set of four processes whose arrival time and burst time are given below :

Process ID	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	5
P4	4	1

Now perform the following if (i) Shortest Job First algorithm and (ii) Preemptive shortest job first (Shortest Remaining Time First) algorithm are used.

- Draw the Gantt chart
- Find average turn-around time
- Find average waiting time
- Find average response time.

- (b) What is the purpose of medium term scheduler?

(4+4)+2

5. (a) When does a page fault occur?

- (b) Consider the page reference stream :

4, 7, 6, 1, 7, 2, 6, 7, 1, 2, 1, 7, 4, 6, 1.

For a system using 3 page frames, find the number of page faults, for the following algorithms :

- (i) First-In-First-Out

- (ii) Optimal page replacement algorithm.

2+(4+4)

6. (a) Distinguish between paging and segmentation.

- (b) Consider a memory of total size 1 MB that follows fixed memory partitioning scheme having partitions 250 KB, 150 KB, 300 KB, 200 KB and 124 KB. If sequence requests for blocks of sizes come in order 145 KB, 100 KB, 190 KB and 290 KB, find the memory allocation and internal fragmentation for each of the following algorithms :

- (i) First Fit

- (ii) Best Fit

- (iii) Worst Fit.

4+(2+2+2)

7. (a) What is a file? State the main features of a file.

- (b) Differentiate between binary semaphore and counting semaphore. Explain how semaphore can be used to solve the readers-writers problem.

(1+3)+(2+4)

8. Write notes on *any two* of the following :

5×2

- (a) Round Robin scheduling

- (b) Belady's Anomaly

- (c) Distributed operating system

- (d) Interprocess Communication.